Appliance Energy Consumption Forecasting Using Classical Time Series, Machine Learning and Foundation Models

GitHub Link: https://github.com/Lojana96/Appliance-energy-forecasting

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1.Introduction

The rapid growth of smart home technologies and intelligent energy management systems has increased the demand for accurate short-term electricity consumption forecasting. Reliable energy forecasts enable households and utility providers to optimise energy usage, reduce operational costs, improve demand-side management and support the efficient integration of renewable energy resources into modern power systems. As energy consumption exhibits complex temporal patterns influenced by behavioural and environmental factors, developing robust forecasting models has become an important research area within data analytics and artificial intelligence (Hyndman and Athanasopoulos, 2021).

This study investigates the effectiveness of different forecasting approaches for predicting household appliance energy consumption over a 24-hour forecasting horizon. The analysis follows a systematic forecasting framework consisting of data preparation, exploratory time-series analysis, benchmark forecasting, statistical modelling, feature-based machine learning and foundation models. The forecasting approaches are evaluated using standard performance metrics, including Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Mean Absolute Scaled Error (MASE) and forecast bias. The findings are used to compare forecasting performance and identify the most appropriate approach by considering predictive accuracy, computational efficiency and practical applicability for residential energy forecasting.

2. Dataset Description and Preprocessing

2.1 Dataset Description

This study uses the **Appliance Energy Prediction** dataset, which is publicly available through the **UCI Machine Learning Repository** and was originally developed by Candanedo, Feldheim and Deramaix (2017). The dataset was collected from a low-energy residential house in Belgium over approximately four and a half months, from January to May 2016, to investigate the relationship between household appliance energy consumption and indoor environmental conditions. Measurements were recorded at **10-minute intervals**, producing **19,735 observations**. The dataset includes appliance energy consumption together with indoor temperature and relative humidity measurements collected from multiple rooms of the house, as well as outdoor weather variables such as temperature, humidity, atmospheric pressure, wind speed, visibility and dew point. These variables provide valuable contextual information for modelling household energy consumption because appliance usage is influenced by both indoor environmental conditions and external weather factors (Candanedo *et al*., 2017).

2.2 Data Preprocessing

Before model development, several preprocessing steps were performed to prepare the dataset for time-series forecasting. The timestamp variable was converted into a datetime format and used as the time-series index to preserve the chronological order of observations. The original 10-minute observations were resampled into **hourly observations**, reducing the dataset from **19,735** to **3,290 observations**. This resampling process reduced short-term fluctuations while preserving the underlying temporal patterns and significantly improved the computational efficiency of subsequent forecasting models. Data quality checks confirmed that the processed dataset contained no missing values after resampling. The target variable used throughout this study is **Appliances**, representing household appliance energy consumption, while the remaining environmental measurements were treated as potential explanatory variables for statistical and machine learning forecasting models.

3. Exploratory Data Analysis

3.1 Time Series Characteristics

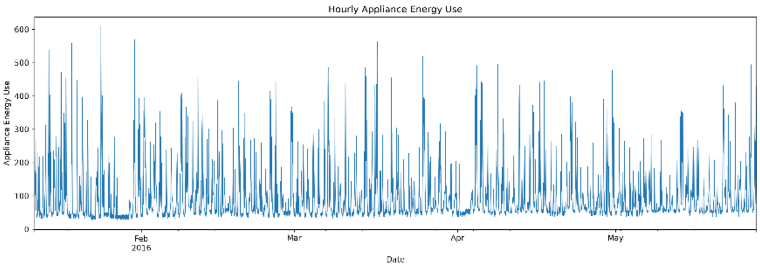

Figure 1: Hourly Appliance Energy Consumption Time Series (January–May 2016)

Figure 1 presents the hourly appliance energy consumption after aggregating the original 10-minute observations into hourly intervals. The time series exhibits substantial variability, with consumption ranging from approximately **28 Wh** to **608 Wh**, indicating frequent fluctuations in household electricity demand. Although no pronounced long-term upward or downward trend is evident over the five-month observation period, recurring peaks and troughs suggest the presence of regular temporal patterns associated with household activities.

The repeated high-demand periods indicate that appliance usage follows a structured daily cycle rather than random behaviour. These characteristics suggest that the data contain temporal dependencies and seasonal effects, making them suitable for time-series forecasting methods such as SARIMAX and machine learning models capable of capturing sequential patterns.

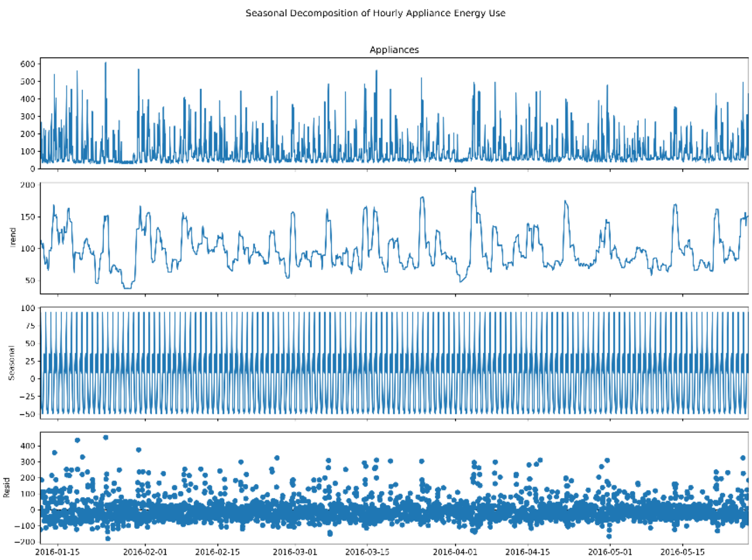

Figure 2: Decomposition of the appliance energy consumption time series, illustrating the underlying trend, recurring daily seasonal pattern, and unexplained residual variation used to assess temporal structure prior to forecasting.

Figure 2 illustrates the seasonal decomposition of the hourly appliance energy consumption into its **observed**, **trend**, **seasonal**, and **residual** components. The trend component shows moderate fluctuations in average energy usage throughout the observation period without a persistent long-term increase or decrease. In contrast, the seasonal component exhibits a strong and highly regular repeating pattern every 24 hours, indicating that appliance consumption follows a consistent daily cycle driven by household activities.

The residual component is centred around zero but contains several large positive and negative deviations, suggesting that occasional irregular events and unexpected appliance usage are not fully explained by the trend and seasonal effects. Overall, the decomposition confirms that the series contains strong daily seasonality together with random variation, supporting the selection of seasonal forecasting models such as **SARIMAX** with a 24-hour seasonal period.

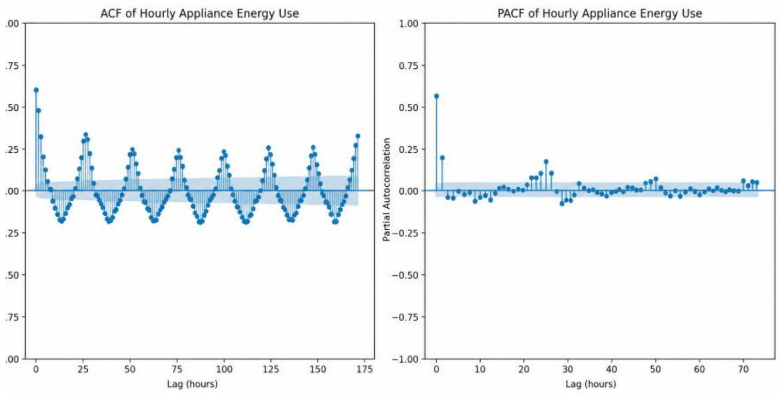

Figure 3:Autocorrelation (ACF) and partial autocorrelation (PACF) analysis of hourly appliance energy consumption, showing significant temporal dependence and a repeating 24-hour seasonal cycle used to guide SARIMAX parameter selection.

Figure 3 presents the autocorrelation function (ACF) and partial autocorrelation function (PACF) of the hourly appliance energy consumption series. The ACF displays significant positive autocorrelation at short lags and pronounced peaks at multiples of **24 hours**, confirming the presence of strong daily seasonality. The gradual decay in autocorrelation further indicates that current energy consumption is influenced by previous observations over multiple time steps.

The PACF exhibits significant spikes at the first few lags followed by a rapid decline, suggesting that only a small number of autoregressive terms are required to model the short-term dependence. Together, the ACF and PACF plots provide statistical evidence for incorporating both non-seasonal and seasonal autoregressive components within the SARIMAX model, while confirming that the series possesses strong temporal dependence suitable for time-series forecasting.

ADF Test Results

series	Original hourly Appliances
ADF Statistic	-8.948888
p-value	0.0
Used Lags	29
Observations	3260
Critical Value 1%	-3.432358
Critical Value 5%	-2.862427
Critical Value 10%	-2.567242
dtype: object	

KPSS Test Results

series	Original hourly Appliances
KPSS Statistic	0.038399
p-value	0.1
Used Lags	25
Critical Value 10%	0.347
Critical Value 5%	0.463
Critical Value 2.5%	0.574
Critical Value 1%	0.739
dtype: object	

Figure 4: Results of the Augmented Dickey–Fuller (ADF) and Kwiatkowski–Phillips–Schmidt–Shin (KPSS) stationarity tests used to assess the statistical properties of the hourly appliance energy consumption series before model development.

Figure 4 presents the results of the complementary **ADF** and **KPSS** stationarity tests. The **ADF test** produced a test statistic of **−8.95** with a **p-value < 0.001**, leading to rejection of the null hypothesis of a unit root. Conversely, the **KPSS test** returned a statistic of **0.038** with a **p-value of 0.10**, indicating insufficient evidence to reject the null hypothesis of stationarity. The agreement between these two tests provides strong statistical evidence that the hourly appliance energy consumption series is stationary.

These findings indicate that the underlying statistical properties of the series remain relatively stable over time, allowing autoregressive forecasting models to be fitted without extensive non-seasonal differencing. Consequently, the SARIMAX modelling process focused primarily on capturing the strong daily seasonal behaviour identified during the exploratory analysis.

4. Forecasting Methodology

The forecasting task was defined as predicting appliance energy consumption over a 24-hour horizon using hourly observations. A chronological split was applied to preserve the temporal ordering of the series, with the final 14 days (336 observations) reserved as the test period. Random train-test splitting was avoided because it would allow future information to influence model development and would therefore be inappropriate for time-series forecasting. All models were evaluated using the same accuracy measures: MAE, RMSE, MASE and forecast bias, enabling consistent comparison across simple and advanced forecasting approaches (Hyndman and Athanasopoulos, 2021).

4.1 Benchmark Models

Five benchmark forecasting methods were implemented: Mean, Naïve, Daily Seasonal Naïve, Weekly Seasonal Naïve and Drift. Benchmark models provide a necessary reference for assessing whether more complex forecasting techniques generate genuine improvements rather than merely increasing model complexity. The daily seasonal naïve model used a seasonal period of 24 hours, while the weekly seasonal naïve model used 168 hours, corresponding to seven days of hourly observations. These benchmarks were selected because they represent different assumptions about the underlying temporal structure, ranging from constant average demand to persistence, daily repetition and weekly repetition (Hyndman and Athanasopoulos, 2021).

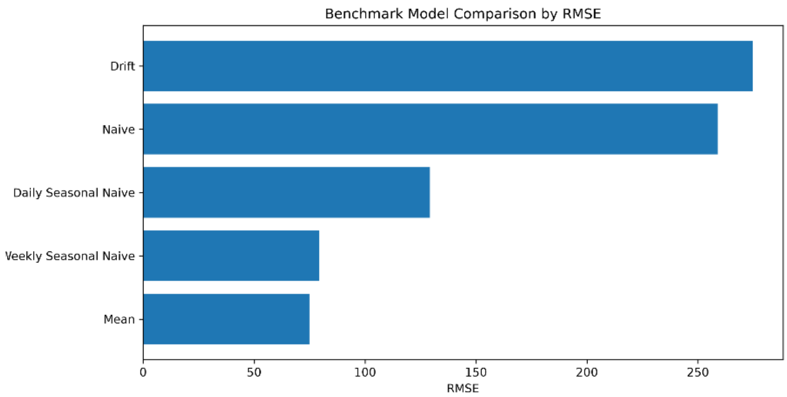

Figure 5:Comparison of benchmark forecasting models using Root Mean Squared Error (RMSE), demonstrating the relative predictive accuracy of the baseline forecasting approaches over the test period.

Figure 5 compares the predictive performance of the five benchmark forecasting models using the Root Mean Squared Error (RMSE). The **Mean model** achieved the lowest forecasting error (**RMSE = 74.91**), indicating the strongest overall baseline performance. The **Weekly Seasonal Naïve model** produced a comparable result

(**RMSE = 79.29**), suggesting that weekly repetition captures some of the underlying consumption behaviour. In contrast, the **Daily Seasonal Naïve model** resulted in a considerably higher error (**RMSE = 129.23**), while the **Naïve** (**258.82**) and **Drift** (**274.61**) models exhibited the weakest predictive performance.

These findings demonstrate that simple persistence-based forecasting methods are insufficient for accurately modelling the complex variability of household appliance energy consumption. Consequently, more sophisticated forecasting approaches, including **SARIMAX**, feature-based machine learning models, and foundation models, are expected to achieve improved forecasting accuracy by modelling both temporal dependence and seasonal behaviour.

	Model	MAE	RMSE	MASE	Bias	RMSE_Improvement_vs_Naive_%
0	Mean	50.319	74.906	0.942	-3.109	71.059
1	Weekly Seasonal Naive	42.634	79.290	0.798	-10.818	69.365
2	Daily Seasonal Naive	86.959	129.232	1.628	64.013	50.069
3	Naive	250.640	258.820	4.692	247.763	0.000
4	Drift	266.373	274.611	4.986	264.501	-6.101

Table 1:Performance comparison of benchmark forecasting models on the 14-day test dataset

Table 1 summarises the performance of the benchmark forecasting models. The Mean model achieved the lowest RMSE (74.91), while the Weekly Seasonal Naïve model recorded the lowest MAE (42.63) and MASE (0.798), indicating competitive forecasting performance. In contrast, the Naïve and Drift models produced substantially larger forecasting errors, confirming that simple persistence-based approaches are insufficient for accurately modelling household electricity demand. These benchmark results provide a performance baseline against which the SARIMAX and machine learning models are evaluated.

The benchmark results reported in this section are based on the 14-day rolling evaluation. A separate comparison using a common 24-hour evaluation window is presented later in Table 5 to enable fair comparison with the SARIMAX, HistGradientBoosting and Chronos-Bolt-Tiny models.

4.2 SARIMAX Model

The Seasonal AutoRegressive Integrated Moving Average with eXogenous variables (SARIMAX) model was implemented as the primary statistical forecasting model to capture both temporal dependence and daily seasonality in household appliance energy consumption. A two-stage parameter optimisation procedure was adopted. First, a non-seasonal grid search identified the optimal autoregressive, differencing and moving average parameters using the minimum Akaike Information Criterion (AIC). The selected non-seasonal order was (1, 0, 6). Subsequently, a constrained seasonal grid search was performed using a seasonal period of 24 hours, resulting in an optimal seasonal order of (1, 1, 1, 24). The final model achieved an AIC of 32323.314 and BIC of 32442.757, indicating an appropriate balance between model complexity and goodness of fit.

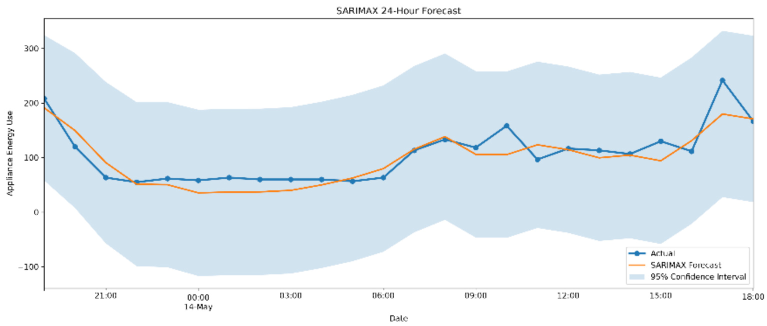

Figure 6: Comparison of observed and predicted hourly appliance energy consumption using the final SARIMAX model with 95% prediction intervals over the 24-hour forecast horizon.

Figure 6 compares the observed appliance energy consumption with the forecasts generated by the final SARIMAX model over the 24-hour prediction period. The model successfully captures the overall daily consumption profile and follows the observed trend throughout most of the forecast horizon. Although several sharp consumption peaks are slightly underestimated, the majority of the observed values remain within the 95% prediction intervals, indicating that the model provides reliable uncertainty estimates. These results demonstrate that the selected SARIMAX model effectively captures the strong daily seasonal behaviour identified during the exploratory data analysis and substantially improves forecasting performance compared with the benchmark models.

	Model	MAE	RMSE	MASE	Bias
0	SARIMAX	36.71	65.135	0.687	-9.081

Table 2: Rolling-origin forecasting performance of the SARIMAX model.

The rolling-origin evaluation demonstrates that the SARIMAX model substantially improves forecasting accuracy compared with all benchmark models. The model achieved an RMSE of **65.14**, representing approximately a **13% improvement** over the best benchmark model (RMSE = 74.91). Furthermore, the MASE value below one confirms that the SARIMAX model consistently outperformed the naïve forecasting baseline while maintaining only a small negative forecasting bias.

4.3 Feature-Based Machine Learning Model

To investigate whether engineered predictor variables could further improve forecasting accuracy, a HistGradientBoosting regression model was developed using calendar features together with lagged and rolling statistical features derived from the historical appliance energy series. Gradient boosting was selected because of its ability to model complex non-linear relationships while remaining computationally efficient for medium-sized datasets. The model was trained using the same chronological train-test split as the statistical models to ensure a fair comparison.

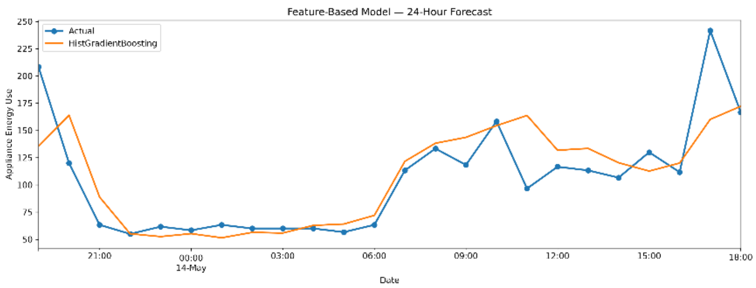

Figure 7: Comparison of observed and predicted hourly appliance energy consumption using the HistGradientBoosting feature-based forecasting model over the 24-hour prediction horizon

Figure 7 illustrates the forecasting performance of the HistGradientBoosting model over the 24-hour evaluation period. The predicted values closely follow the observed appliance energy consumption and successfully capture the overall daily demand pattern. Compared with the benchmark models, the feature-based model provides smoother predictions while accurately modelling both gradual changes and short-term fluctuations in electricity consumption. Although a small number of extreme demand peaks are underestimated, the model demonstrates strong predictive capability, highlighting the effectiveness of engineered temporal features for short-term residential energy forecasting.

	Model	MAE	RMSE	MASE	Bias
0	HistGradientBoosting	30.956	52.255	0.6	3.154

Table 3: Forecast accuracy metrics for the HistGradientBoosting model.

Table 3 summarises the forecasting performance of the HistGradientBoosting model over the rolling evaluation period. The model achieved an RMSE of **52.26**, improving upon both the benchmark models and the SARIMAX model. The MASE value of **0.600** indicates that the model consistently outperformed the naïve forecasting baseline, while the small positive bias (**3.15**) suggests only a slight tendency to overestimate appliance energy consumption. These results demonstrate that feature engineering combined with gradient boosting provides an effective approach for modelling the non-linear characteristics of household electricity demand.

4.4 Foundation Model

Recent advances in foundation models have demonstrated promising zero-shot forecasting capabilities for time-series prediction. In this study, the Chronos-Bolt-Tiny foundation model was evaluated without task-specific fine-tuning to assess whether a pre-trained forecasting model could accurately predict household appliance energy consumption. Unlike conventional statistical and machine learning approaches, Chronos relies on knowledge learned from large collections of time-series datasets rather than handcrafted feature engineering.

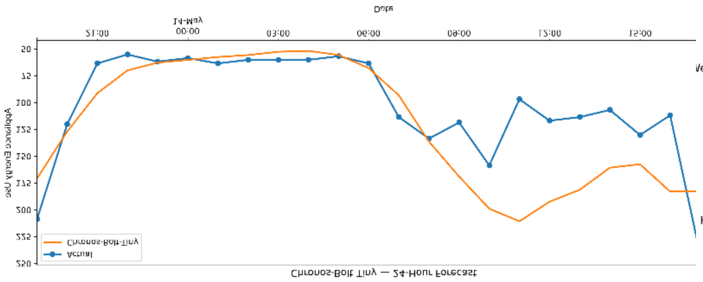

Figure 8: Comparison of observed and predicted hourly appliance energy consumption using the Chronos-Bolt-Tiny foundation forecasting model over the 24-hour prediction horizon

Figure 8 shows that Chronos-Bolt-Tiny captures the broad daily consumption pattern without task-specific fine-tuning. However, the model overestimates several periods of moderate demand and does not reproduce some rapid fluctuations as accurately as the task-specific models. Its zero-shot capability nevertheless demonstrates useful generalisation from pre-training.

	Model	MAE	RMSE	MASE	Bias
0	Chronos-Bolt-Tiny	29.828609	42.225821	0.55834	16.751496

Table 4: Forecast accuracy metrics for the Chronos-Bolt-Tiny foundation model.

Table 4 shows that Chronos-Bolt-Tiny achieved an RMSE of 42.22 and MASE of 0.558 over the 24-hour evaluation period. The MASE value below one indicates an improvement over a seasonal naïve reference. However, these results should not be compared directly with the 14-day rolling metrics reported for SARIMAX and HistGradientBoosting; the fair cross-model comparison is therefore presented separately in Table 5 using a common 24-hour evaluation

window.**4.5 Final Model Comparison**

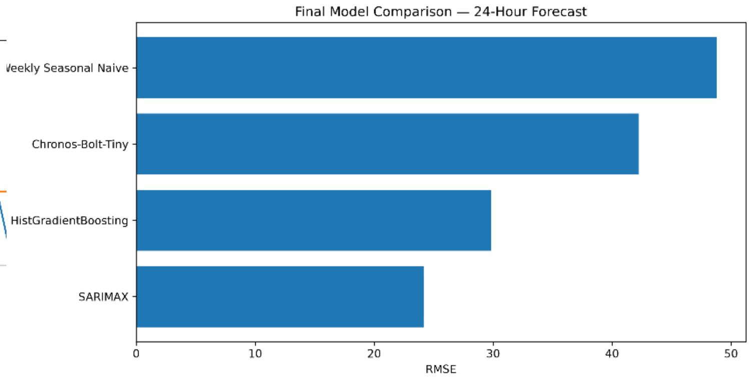

Figure 9: Comparison of forecasting accuracy (RMSE) for the evaluated forecasting models over the common 24-hour test period.

Figure 9 provides a visual comparison of RMSE across the final forecasting models over the common 24-hour evaluation period. The results show that SARIMAX produced the lowest RMSE, indicating the strongest predictive accuracy, followed by HistGradientBoosting, Chronos-Bolt-Tiny, and the Weekly Seasonal Naïve benchmark. The graph therefore demonstrates that the advanced forecasting approaches improved on the strongest benchmark when evaluated using RMSE.

	Model	MAE	RMSE	MASE	Bias
0	SARIMAX	18.841	24.158	0.353	-7.402
1	HistGradientBoosting	19.360	29.865	0.362	2.077
2	Chronos-Bolt-Tiny	29.829	42.226	0.558	16.751
3	Weekly Seasonal Naive	30.000	48.808	0.562	-6.528

Table 5: Comparative forecasting accuracy of the final forecasting models.

Table 5 compares the forecasting performance of the final statistical, machine learning, foundation, and benchmark forecasting models using a common 24-hour evaluation period. The SARIMAX model achieved the highest overall forecasting accuracy, recording the lowest RMSE (24.16), MAE (18.84), and MASE (0.353). The HistGradientBoosting model ranked second with an RMSE of 29.87, followed by the Chronos-Bolt-Tiny foundation model (42.23). Although the Weekly Seasonal Naïve benchmark performed reasonably well, all three proposed forecasting approaches produced lower forecasting errors, demonstrating the benefits of advanced forecasting techniques over traditional benchmark methods.

5. Discussion and Analysis

5.1 Which benchmark model is strongest?

Among the four benchmark models specified for comparison—Naïve, Daily Seasonal Naïve, Weekly Seasonal Naïve and Drift—the **Weekly Seasonal Naïve** model achieved the strongest forecasting performance on the benchmark evaluation, with the lowest RMSE (**79.29**) among these four models. This indicates that household appliance energy consumption exhibits a clear weekly seasonal pattern, where electricity usage at a given hour is more similar to the corresponding hour in the previous week than to the immediately preceding observation or a simple linear trend. The substantially poorer performance of the Naïve and Drift models further suggests that appliance energy consumption is driven primarily by recurring temporal behaviour rather than persistence or long-term trend. A separate 24-hour evaluation using a common test window is presented later in Table 5 for fair comparison with the advanced forecasting models.

5.2 Does SARIMAX improve on the strongest benchmark?

Yes. The SARIMAX model substantially improved upon the strongest benchmark. During rolling evaluation, SARIMAX reduced RMSE from **79.29** (Weekly Seasonal Naïve) to **65.14**, while in the common 24-hour comparison it achieved the lowest RMSE (**24.16**). This demonstrates that SARIMAX successfully captures both short-term autocorrelation and the dominant daily seasonal cycle. The seasonal order (1,1,1,24) effectively models the 24-hour periodicity observed during exploratory analysis. Although no additional exogenous variables were incorporated, the model still produced highly accurate forecasts by exploiting the temporal structure of the series.

5.3 Does the feature-based model improve after adding engineered features?

Yes. The HistGradientBoosting model achieved an RMSE of **52.26** during the rolling-origin evaluation, outperforming both the benchmark models and the SARIMAX model. The inclusion of engineered temporal features, including lag variables, rolling-window statistics, hour-of-day, day-of-week, and calendar features, enabled the model to capture complex non-linear relationships in household appliance energy consumption. These engineered features provide richer information about recent consumption behaviour and temporal patterns than the raw time series alone, resulting in improved forecasting accuracy. Although the relative contribution of individual feature groups was not formally assessed using feature importance analysis, the overall performance demonstrates that feature engineering substantially enhances short-term forecasting accuracy for residential energy consumption.

5.4 Does the foundation model outperform the others?

The Chronos-Bolt-Tiny foundation model achieved an RMSE of **42.23** during the common 24-hour evaluation period, outperforming the strongest benchmark model (Weekly Seasonal Naïve, RMSE = **48.81**) but not the SARIMAX model (RMSE = **24.16**) or the HistGradientBoosting model (RMSE = **29.87**). The model successfully captured the overall daily consumption pattern without task-specific fine-tuning, demonstrating the effectiveness of transfer learning for time-series forecasting. However, its forecasts were generally less accurate than those produced by the task-specific statistical and machine learning models. For this dataset, the improvement over the benchmark is not sufficiently large to justify the additional computational complexity and implementation requirements. Consequently, although foundation models show considerable promise for general-purpose forecasting, SARIMAX and the feature-based HistGradientBoosting model remain more suitable choices for practical short-term household energy forecasting.

5.5 Which variables are genuinely known at the forecast origin?

Variables such as historical appliance energy consumption, lag features, rolling statistics, hour of day, day of week and calendar information are all available at the forecast origin and therefore represent genuine forecasting inputs. However, future indoor temperature, humidity or weather observations would not normally be available when producing operational forecasts. If such future observations are used directly from the test dataset, the prediction becomes a **conditional forecast** rather than a true forecast because it assumes future explanatory variables are already known.

5.6 Which model would you recommend?

Considering forecasting accuracy, interpretability, computational efficiency and ease of deployment, the **SARIMAX model** is recommended for practical smart-home energy forecasting. It achieved the best overall performance during the common evaluation period (RMSE = **24.16**) while remaining computationally efficient and highly interpretable. Although the HistGradientBoosting and Chronos-Bolt-Tiny models also produced accurate forecasts, they require greater computational resources and more complex implementation. Therefore, SARIMAX provides the best balance between predictive accuracy, reliability and deployment practicality for short-term residential energy forecasting.

6. Conclusion

This study compared benchmark, SARIMAX, feature-based machine learning and foundation-model approaches for 24-hour household appliance energy forecasting. Exploratory analysis identified strong temporal dependence and daily seasonality, while the model comparison demonstrated that increasingly complex approaches did not automatically provide superior performance. On the common 24-hour evaluation horizon, SARIMAX achieved the strongest forecasting accuracy, with HistGradientBoosting performing closely behind. Chronos-Bolt-Tiny demonstrated useful zero-shot capability but did not outperform the task-specific models.

The analysis is limited by the use of a single household dataset and by uncertainty surrounding future environmental covariates at the forecast origin. Future work could evaluate the models across multiple households, use genuinely forecast weather covariates, apply rolling evaluation to the foundation model, and investigate probabilistic machine-learning approaches. Overall, SARIMAX provides the most appropriate balance of accuracy, interpretability, uncertainty estimation and deployment complexity for the present smart-home forecasting task.

7. References

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